Damilola Ogunleye

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Education

- **University College London** — London, UK
  Master of Engineering in Robotics and Artificial Intelligence — *2023 - 2027*
  Overall Average: *80.43% (1st Class)*
  Modules: *Scene Understanding and Situational Awareness, Robot Learning, Deep Learning for Robotics and Artificial Intelligence*
- **St Olav VGS** — Stavanger, Norway
  International Baccalaureate Diploma Programme — *2021 - 2023*
  Grade: *41/45 Maths AA HL: 7 Physics HL: 7*

Skills Summary

- **Languages & Tools**: Python, PyTorch, NumPy, ROS2, C, scikit-learn, Git, Gazebo, PyBullet, MuJoCo

Experience

- **HyBird** — London, UK
  Research Engineer (Part Time) — *Sep 2026 - Present*
  - Developing a multimodal AI pipeline for 3D spatial understanding, benchmarking emerging models against conventional methods through reproducible evaluations and analysing failure modes to assess robustness and consistency in large-scale environments and across input modalities.
- **Google DeepMind Research Ready Programme** — York, UK
  Research Assistant Intern — *Jun 2026 – Aug 2026*
  - Developed a ROS 2 Jazzy and Gazebo Classic simulation environment to evaluate decentralised swarm coordination for autonomous nuclear waste clustering, implementing the CacheConsensus algorithm.
  - Integrated and validated the robotics autonomy stack across simulation and physical platforms, assessing consistency between simulated and real-world behaviour.
  - Built a computer vision dataset for nuclear waste classification and localisation, guided a teammate through model fine-tuning, and investigated failure cases across simulation and physical testing; achieved 90% precision and 0.022 m RMSE localisation error while applying AMLAS principles to support safety assurance (Repository)
- **National Centre for Artificial Intelligence and Robotics** — Abuja, Nigeria
  Robotics Intern — *Aug 2025 – Sep 2025*
  - Designed and prototyped an autonomous agricultural robot featuring Wi-Fi-enabled monitoring, a seed-dispensing mechanism, and an automated furrow-opening system using an Arduino-based embedded controller.
  - Implemented GPS, IMU, and wheel encoder sensor fusion using a Kalman filter, tuning measurement and process noise covariance matrices to improve localisation accuracy and navigation robustness during autonomous operation.
- **Cisco (Outshift by Cisco)** — London, UK
  Agentic AI Engineering Intern — *Apr 2025 – Jul 2025*
  - Designed and deployed a multi-agent AI workflow using LangChain and LangGraph to generate, critique and iteratively refine CAD designs from natural-language specifications, integrating the workflow with automated 3D printing.

Projects

- **UKSEDS Olympus Rover Trials (Advanced Stream) (November 2025 - July 2026)**: Perception, Navigation and Simulation Lead. Led development of a ROS 2 rover autonomy stack integrating RGB-D/LiDAR SLAM, A* path planning and MPPI control, optimised for deployment on Jetson and Raspberry Pi hardware.
- **Reinforcement Learning Control of Efficient Soft Exoskeletons (September 2025 - June 2026)**: Developed a PPO-based controller for a soft exoskeleton, optimising sit-to-stand transitions for completion time, stability and adherence to natural joint trajectories; achieved assistance equivalent to a 10% reduction in muscle activation, evaluated against unassisted movement in simulation and supported by physical experiments using EMG sensors.
- **Multimodal AI Literacy Agent – Lead Deep Learning Researcher (September 2025 - December 2025)**: Led a cross-functional team developing a multimodal AI agent, integrating an LLM with LoRA fine-tuning on two datasets, RAG and a YOLO/CNN vision pipeline for real-time interaction; investigated fine-tuning strategies and RAG configurations, evaluating language-model performance using perplexity and benchmarking against commercial agents.
- **A Stochastic Simulation Framework for Robotic Grasp Stability Classification. (October 2024 - December 2024)**: Developed an object-oriented PyBullet simulation to stochastically generate grasping datasets, extracting kinematic and contact features to train PyTorch MLP and scikit-learn SVM classifiers, achieving 79% accuracy on a held-out test set for grasp stability prediction.

Honors and Awards

- Best Innovation Award & 2nd Place Overall – UKSEDS Olympus Rover Trials (2025)
- **1st Place** – Google BGN Hackathon (Spotlight by Gemi9), Oct. 2024

Extracurriculars

- Clarinettist with the All Souls Orchestra; former member of the Rogaland Regional Youth Symphony Orchestra.
- Member of the UCL AI Society, Computer Science Society, and Jazz Society.